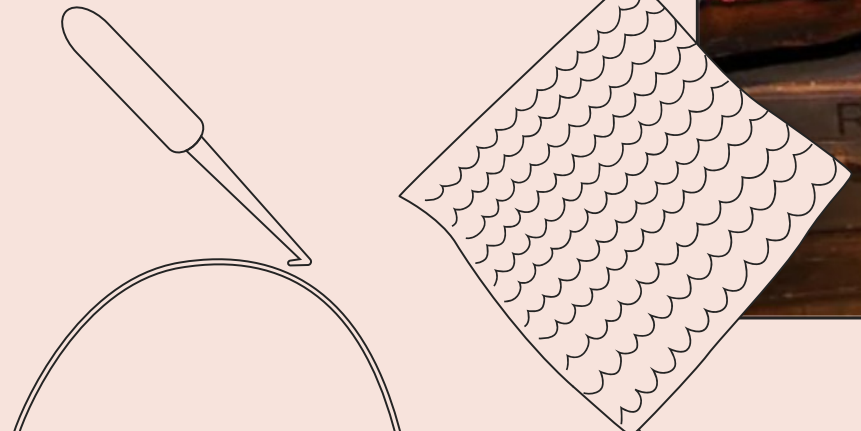
Granny Square Afghan Analysis Proposal

By Andi Shievitz









01.

Topic

- Granny Square Blanket
- Experimental Units: Squares
- Coded A1-J8 (80 Squares)







03.

Data Collection

- Goal: 1 Square/Day
- Timing each Square (Stopwatch)
- Measuring Length of Yarn Used/Square (Line Counter)
- Measuring Pre-Blocked Squares (Caliper)

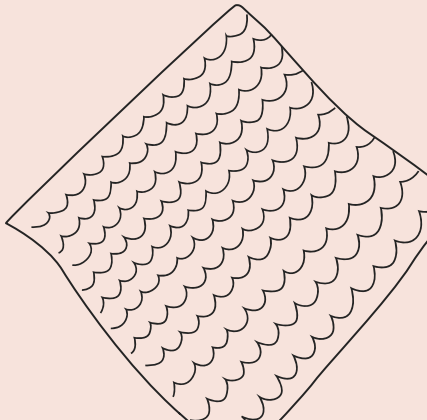
Data:

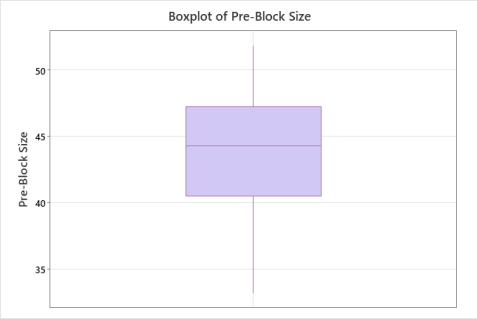
Square Code – Unique, PK
Direction – Categorical, Binary
Pattern - Categorical
Completed – Numeric, Binary
Time – Continuous
of Sts – Numeric, Binary
Length - Continuous
Pre-Block Size - Continuous
$/Sts - Continuous
Opinion – Categorical, Binary

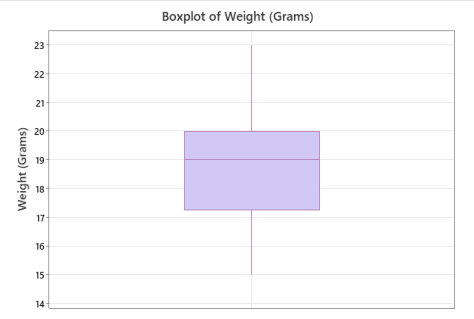
of Colors - Continuous
PrimaryColor - Categorical
SecondaryColor – Categorical
TertiaryColor - Categorical
QuaternaryColor - Categorical
QuinaryColor - Categorical
PCBrand - Categorical
SCBrand – Categorical
TCBrand - Categorical
QACBrand - Categorical
QICBrand - Categorical

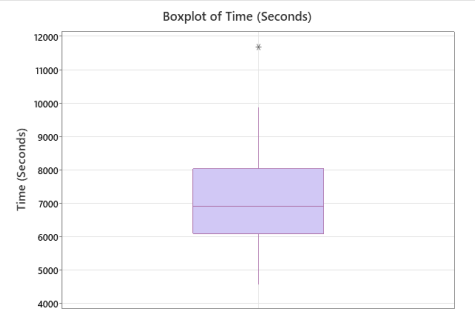
04.

Variables

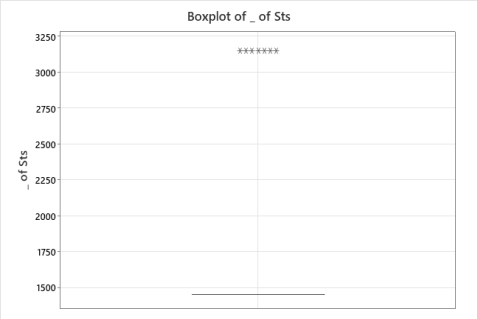


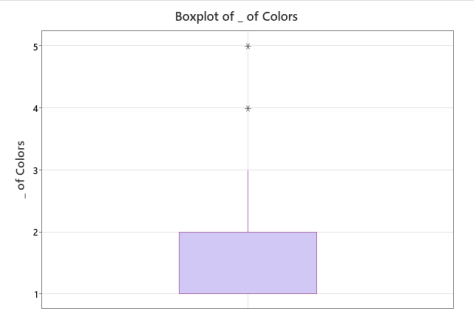






1H
58Min
14Sec

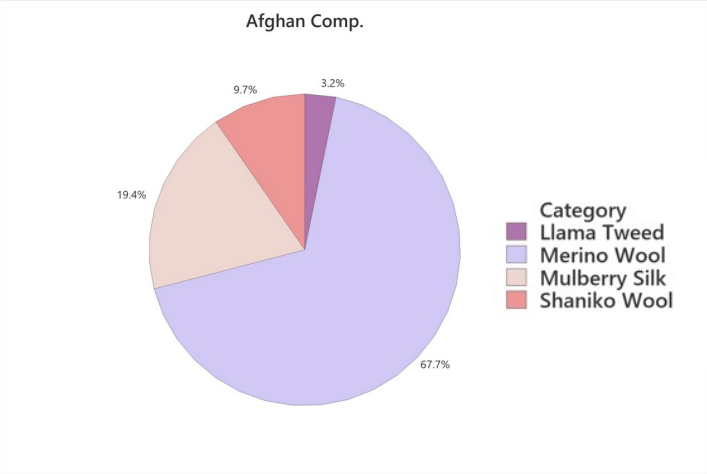


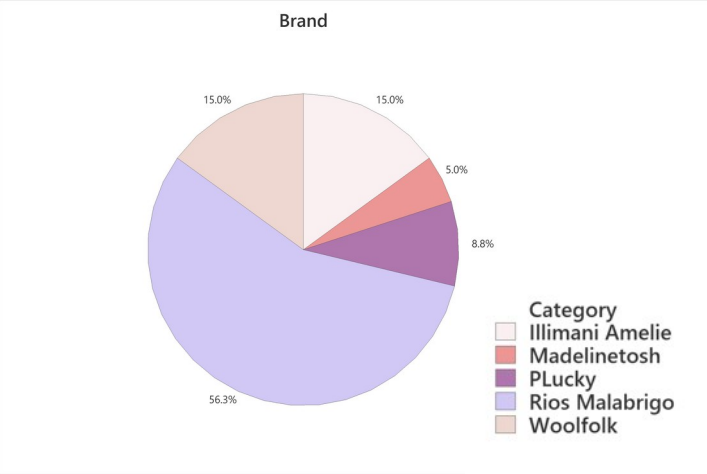


Descriptive Stats

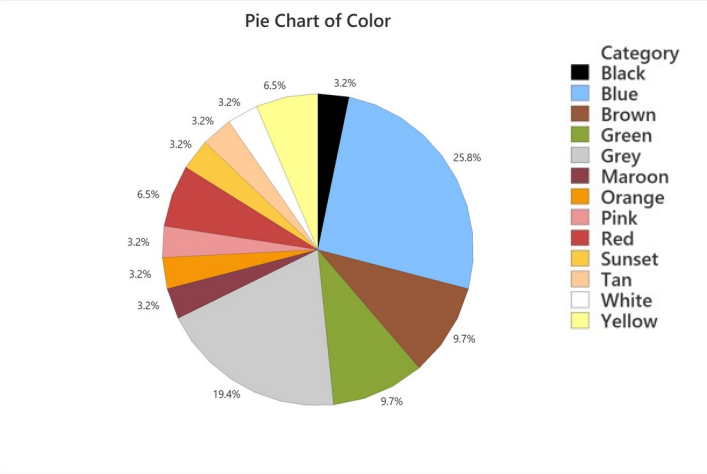
Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum			
Time (Seconds)	80	0	7094.98	159.826	1429.53	4560.02	6090.06	6905.89	8027.21	11679.5			
_ of Colors	80	0	1.575	0.0810825	0.725224	1	1	1	2	5			
_ of Sts	80	0	1602.4	53.9179	482.256	1454	1454	1454	1454	3150			
Weight (Grams)	80	0	18.75	0.177447	1.58713	15	17.25	19	20	23			
Pre-Block Size	80	0	43.4695	0.539942	4.82938	33.2146			40.5260	44.3105	47.2674	51.8343	





06.



Categorical Variables of Interest

Do squares take longer than 5400 seconds on average to complete?

Test

Null hypothesis H_o: μ = 5400

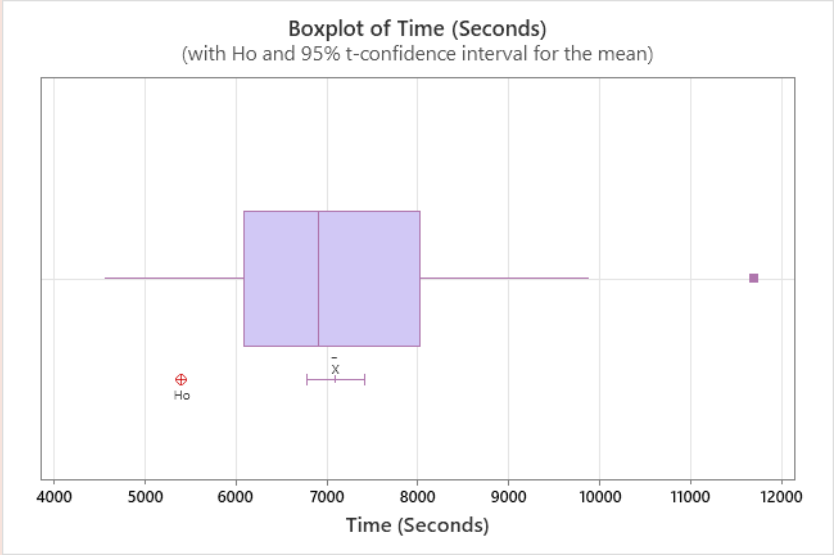
Alternative hypothesis H_1: μ ≠ 5400

T-Value	P-Value
10.61	0.000

Descriptive Statistics

N	Mean	StDev	SE Mean
80	7095 1H 58Min 14Sec	1430	160

95% CI for μ
(6777, 7413)
μ: population mean of Time (Seconds)

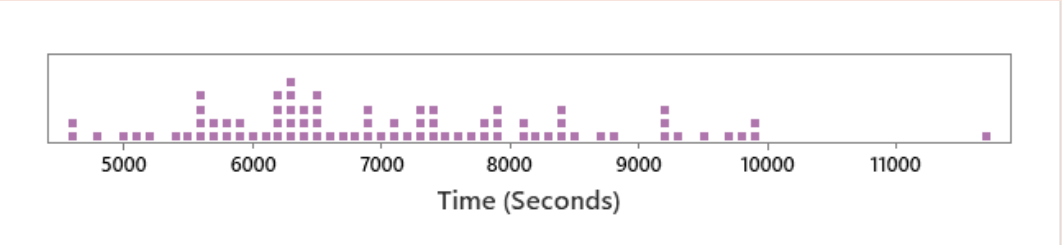


One Sample T-Test

Is the median time to knit a square different from 5400 seconds?

Descriptive Statistics

Sample	N	Median	
Time (Seconds)	80	6	
	1H 55Min 6Sec		



Test

Null hypothesis H_o: $\eta = 5400$

Alternative hypothesis H_1: $\eta \neq 5400$

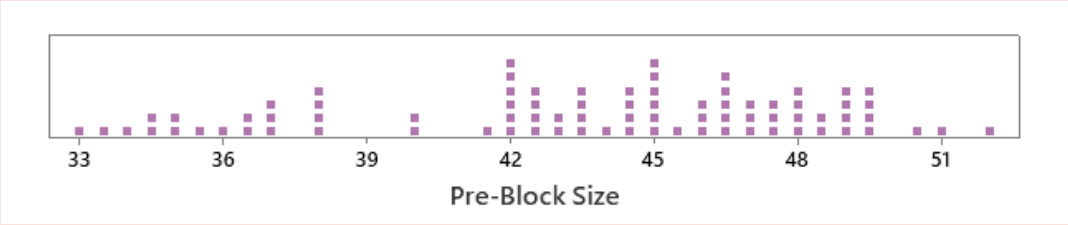
Sample	Number < 5400	Number = 5400	Number > 5400	P-Value
Time (Seconds)	6	0	74	
	0.000			

Sign Test

Is the median pre-block size of a square different from 49 inches squared?

Descriptive Statistics

Sample	N	Median
Pre-Block Size	80	43.7562



Test

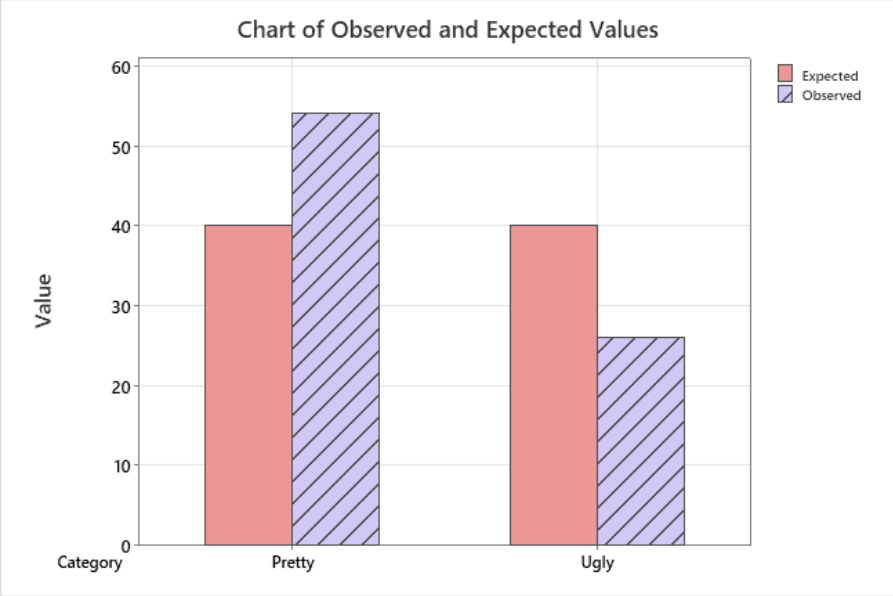
Null hypothesis	H_0: η = 49
Alternative hypothesis	H_1: η ≠ 49

Sample	N for Test	Wilcoxon	P-Value
Pre-Block Size	80	96.00	0.000

Signed Rank Statistic of 96 summarizes how far the ranked differences of my 80 squares are from the hypothesized median of 49 square inches; p value shows strong evidence the true median is not 49.

Signed Rank Test

Are Pretty vs Ugly squares equally common?



Chi-Square Test

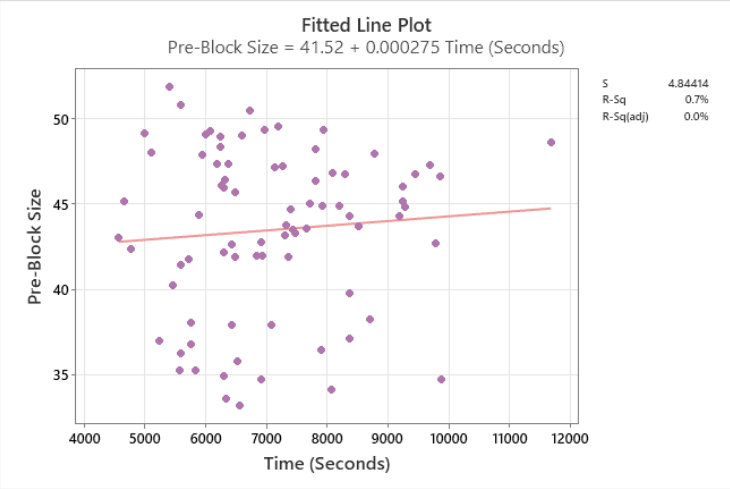
N	DF	Chi-Sq	P-Value
80	1	9.8	0.002
		Larger, values differ greatly	

Observed and Expected Counts

Category	Observed	Test Proportion	Expected	Contribution to Chi-Square
Pretty	54	0.5	40	4.9
Ugly	26	0.5	40	4.9

One-Way Chi Squared

Does the time needed to crochet a square impact its pre-blocked size?



Pre-Block Size = 41.52 + 0.000275 Seconds

Analysis of Variance

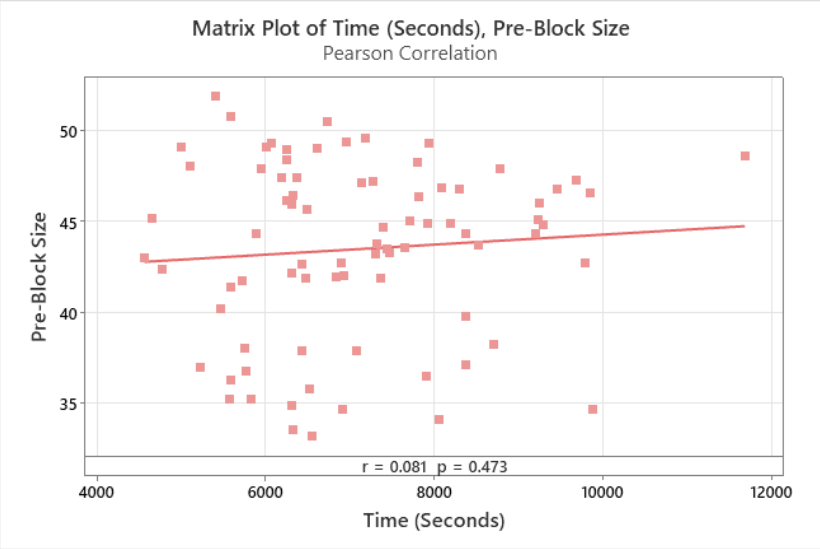
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	12.19	12.19	0.52	0.473
Time (Seconds)	1	12.19	12.19	0.52	0.473
Error	78	1830.32	23.47		
Total	79	1842.51			

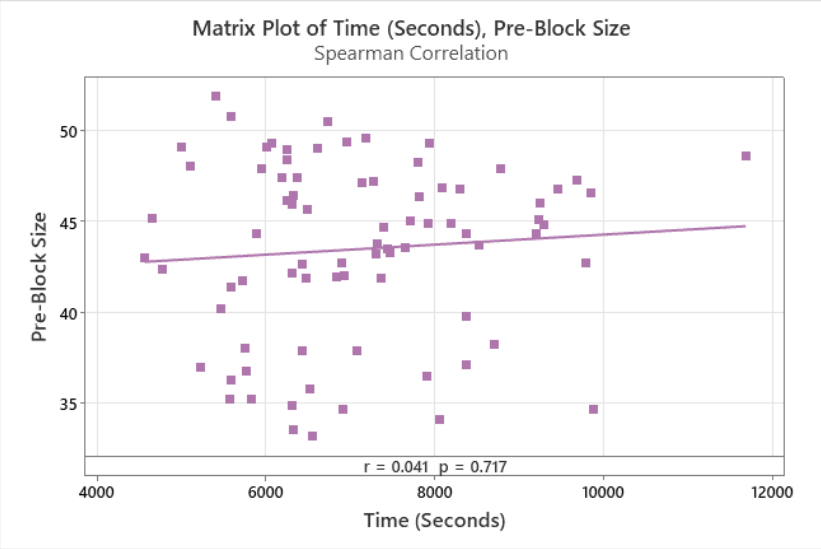
Model Summary

S	R-sq
4.84414	0.66%

Simple Linear Regression

Pearson vs Spearman Correlation





Pearson Correlation

(Seconds)	Time
Pre-Block Size	0.081

Spearman Correlation

(Seconds)	Time
Pre-Block Size	0.041

Pearson vs Spearman

Is pattern type related to whether a square is Pretty or Ugly?

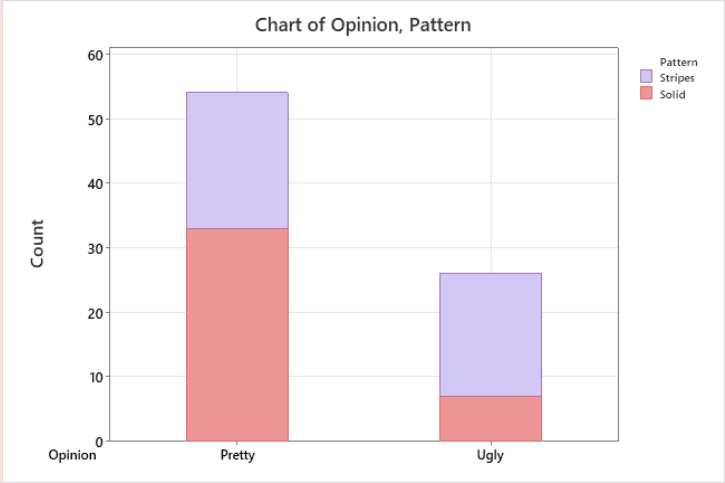
Chi-Square Test

	Chi-Square	DF	P-Value
Pearson	8.205	1	0.004
Likelihood Ratio	8.443	1	0.004

Rows: Pattern Columns: Opinion

	Pretty	Ugly	All
Solid	33	7	40
Stripes	21	19	40
All	54	26	80

Cell Contents
Count



Two-Way Chi Squared

Do diagonal squares take longer to knit than horizontal squares?

Descriptive Statistics: Time (Seconds)

Direction	N	Mean	StDev	SE Mean
Diagonal	7	9944	796	301
Horizontal	73	6822	1151	135

Estimation for Difference

Difference	95% CI for Difference
3122	(2362, 3882)

Test

Null hypothesis		$H_0: \mu_1 - \mu_2 = 0$
Alternative hypothesis		$H_1: \mu_1 - \mu_2 \neq 0$
T-Value	**DF**	**P-Value**
9.48	8	0.000



Two Sample T-Test

Do Pretty squares take longer to knit than Ugly ones?

Method

η_1: median of Pretty
η_2: median of Ugly
Difference: $\eta_1 - \eta_2$

Descriptive Statistics

Sample	N	Median
Pretty	54	6540.09
Ugly	26	7238.54

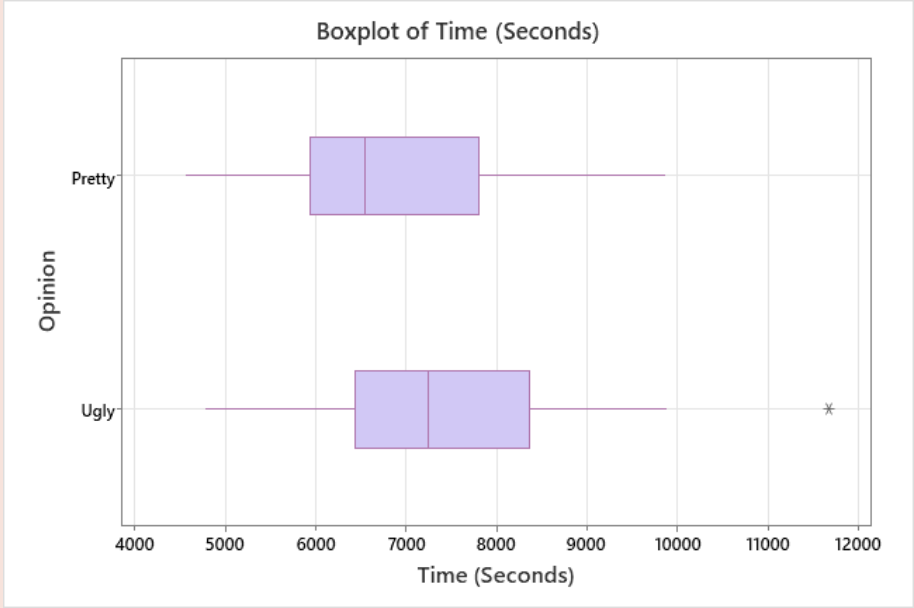
Estimation for Difference

Difference	CI for Difference	Achieved Confidence
-535.184	(-1151.66, 164.63)	95.08%

Test

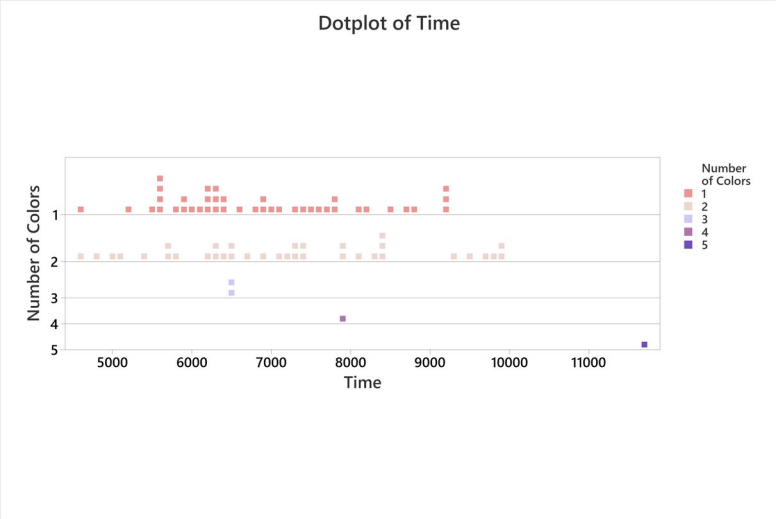
Null hypothesis H_0: $\eta_1 - \eta_2 = 0$
Alternative hypothesis H_1: $\eta_1 - \eta_2 \neq 0$

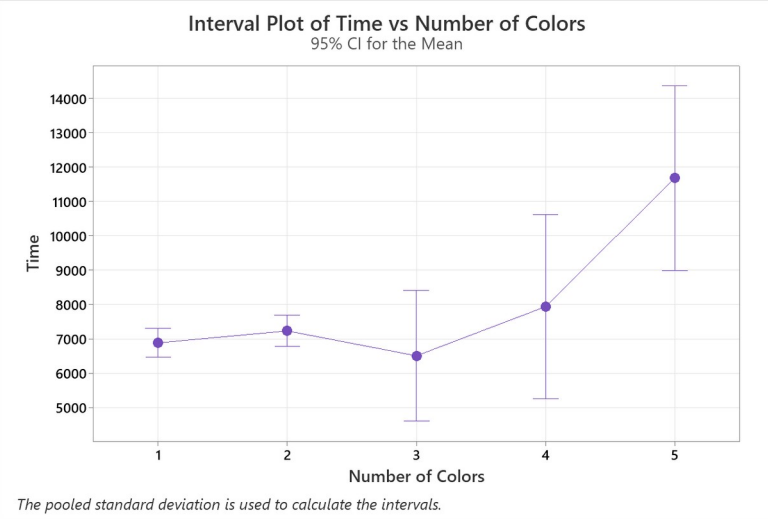
W-Value	P-Value
2035.00	0.120



Rank Sum Test

Does number of colors affect knitting time?



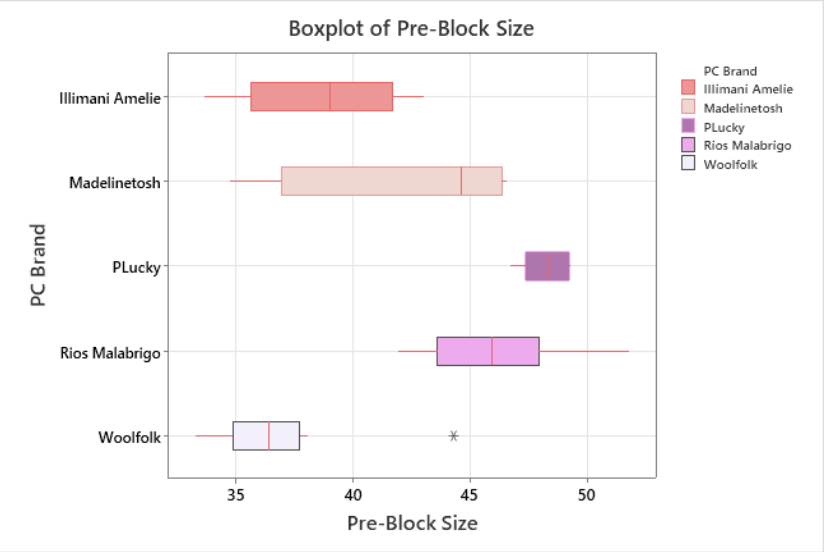


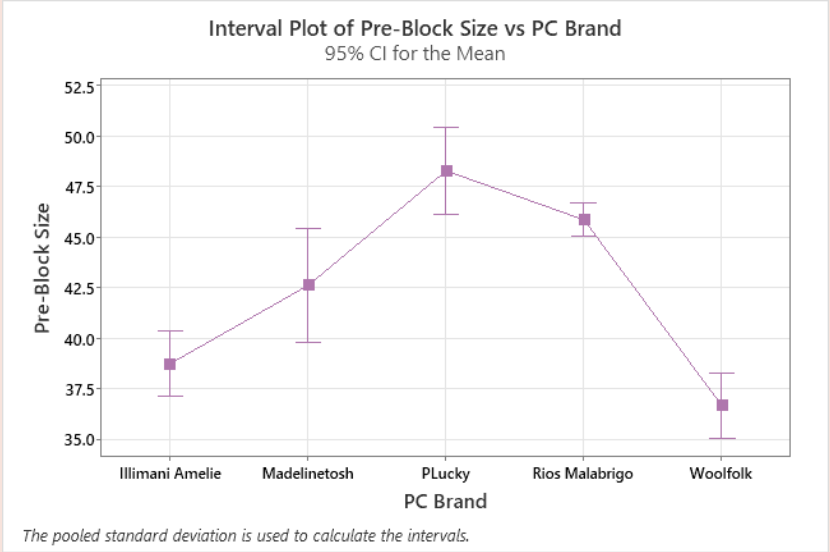
Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Number of Colors	4	24963068	6240767	3.43	0.012
Error	75	136477886	1819705		
Total	79	161440953			

ANOVA

Does the brand of yarn affect pre-blocking size?





Analysis of Variance

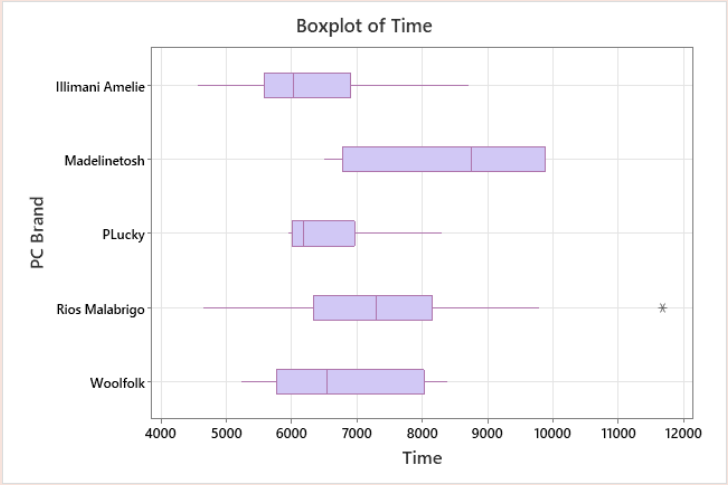
Source	DF	Adj SS	Adj MS	F-Value	P-Value
PC Brand	4	1242.2	310.539	38.79	0.000
Error	75	600.4	8.005		
Total	79	1842.5			

ANOVA

Does primary yarn brand affect knitting time?

Descriptive Statistics

PCBrand	N	Median	Mean Rank	Z-Value
Illimani Amelie	12	6030.05	26.4	-2.28
Madelinetosh	4	8750.24	61.3	1.83
PLucky	7	6180.04	29.6	-1.30
Rios Malabrigo	45	7297.05	45.1	1.99
Woolfolk	12	6535.33	36.9	-0.58
Overall	80		40.5	



Test

Null hypothesis H_0: All medians are equal

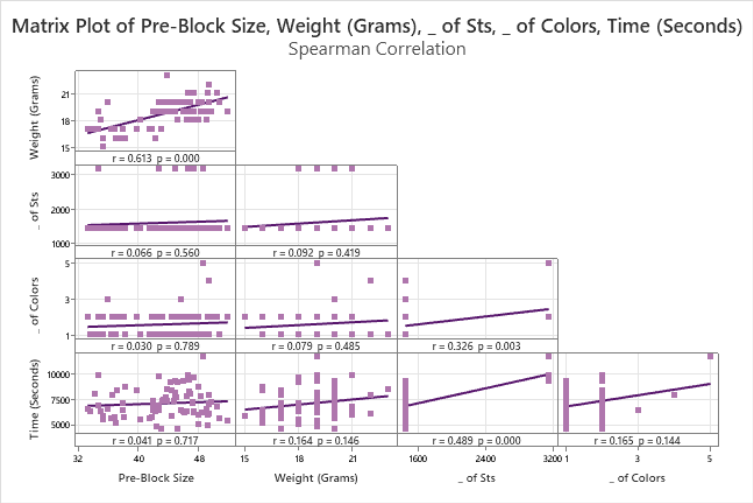
Alternative hypothesis H_1: At least one median is different

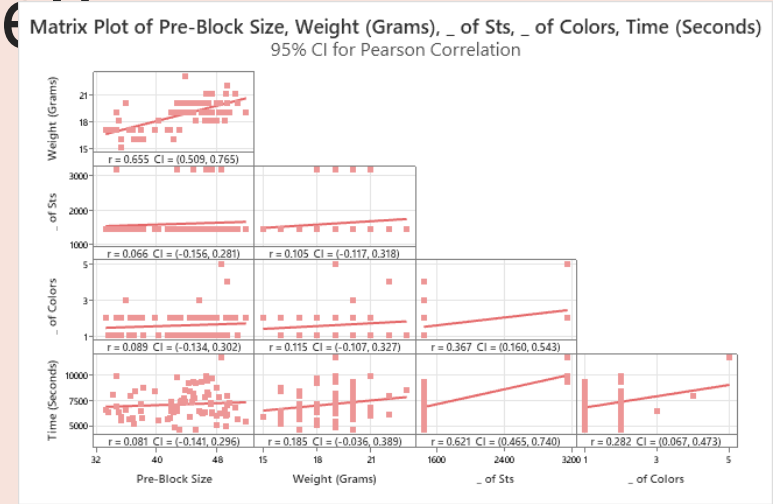
DF	H-Value	P-Value
4	11.17	0.025

The chi-square approximation may not be accurate when some sample sizes are less than 5.

Kruskal-Wallis

What factors best predict knitting





Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	65417392	16354348	12.77	0.000
_ of Colors	1	400669	400669	0.31	0.578
_ of Sts	1	48206950	48206950	37.65	0.000
Weight (Grams)	1	2381433	2381433	1.86	0.177
Pre-Block Size	1	425038	425038	0.33	0.566
Error	75	96023561	1280314		
Total	79	161440953			

Multiple Linear

Regression Equation

Time = 2277 + 106 # of Colors + 1.745 # of Sts + 146 Weight - 20.1 Pre-Block Size



$642.55

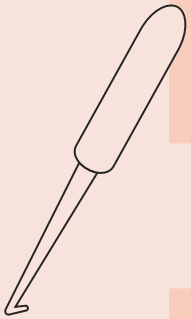
Total Price of Yarn

$3037.28

Total price if made by a minimum wage worker

157h 40m 12s

Crocheting Time (not including sewing)





132,822

Stiches

386,000 inches

Yarn Used (0.3 Miles)